



Digital Integrated Circuit Design

Lecture-3: Number Systems and Basic Arithmetic

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Outline

☐ **Number Systems**

- **Sign-Magnitude Representation**
- **2's and 1's Complement Representation**
- **Floating Point Number**
- **Signed Digit Number**

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The Scope of Computer Arithmetic

Hardware

Design of efficient digital circuits for primitive and other arithmetic operations such as $+$, $-$, $\times$, $\div$, $\sqrt{}$, $\log$, $\sin$, and $\cos$

Issues: Algorithms
Error analysis
Speed/cost trade-offs
Hardware implementation
Testing, verification

Software

Numerical methods for solving systems of linear equations, partial differential equations and so on

Issues: Algorithms
Error analysis
Computational complexity
Programming
Testing, verification

General-Purpose

Flexible data paths
Fast primitive operations like $+$, $-$, $\times$, $\div$, $\sqrt{}$
Benchmarking

Special-Purpose

Tailored to application areas such as
Digital filtering
Image processing
Radar tracking

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Example for Computer Arithmetic

❑ Using a calculator with $\sqrt{}$, x^2 , and x^y functions, compute:

$$\begin{aligned} \blacksquare u &= \sqrt{\sqrt{\dots \sqrt{2}}} = 1.000\ 677\ 131 && \text{"1024}^{\text{th}} \text{ root of 2"} \\ \blacksquare v &= 2^{1/1024} = 1.000\ 677\ 131 \end{aligned}$$

❑ Save u and v ; If you can't save, you have to recompute the values when needed.

$$\begin{aligned} \blacksquare x &= (((u^2)^2)\dots)^2 = 1.999\ 999\ 963 \\ \blacksquare x' &= u^{1024} = 1.999\ 999\ 973 \\ \blacksquare y &= (((v^2)^2)\dots)^2 = 1.999\ 999\ 983 \\ \blacksquare y' &= v^{1024} = 1.999\ 999\ 994 \end{aligned}$$

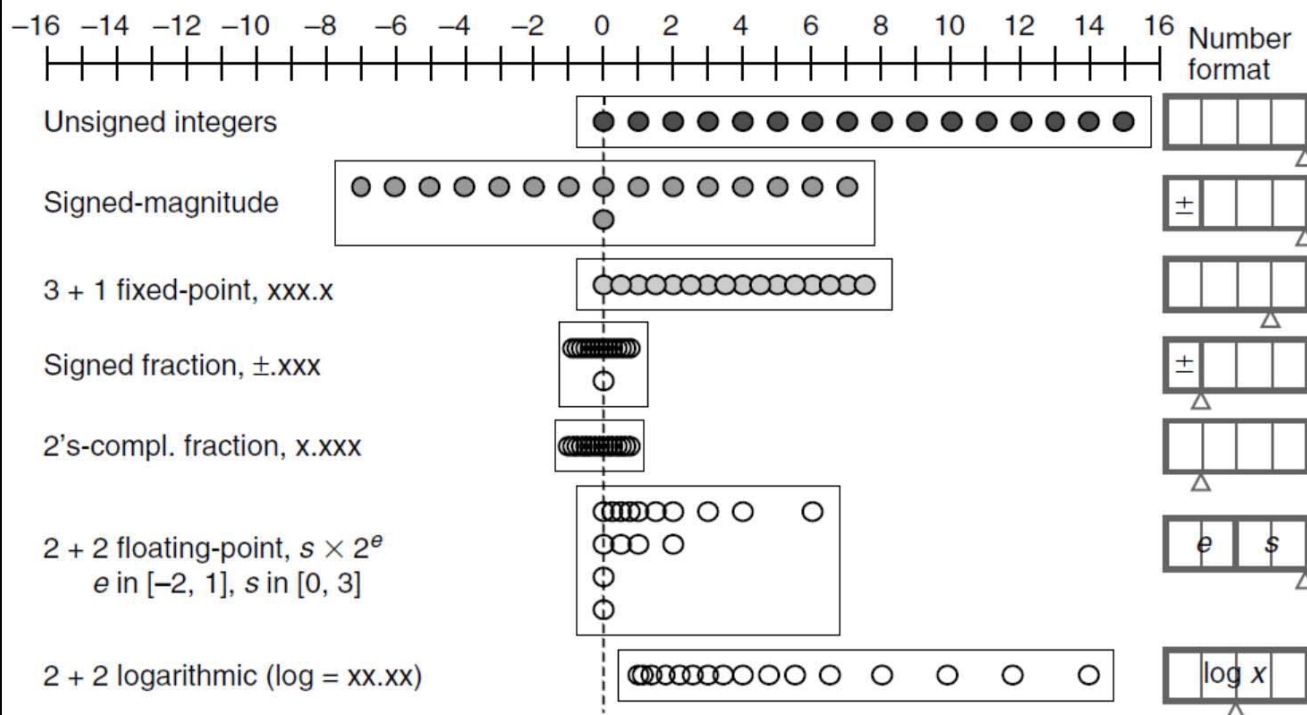
❑ Perhaps v and u are not really the same value.

$$\begin{aligned} \blacksquare w &= v - u = 1 \times 10^{-11} && \text{Nonzero due to hidden digits} \\ \blacksquare (u - 1) \times 1000 &= 0.677\ 130\ 680 && [\text{Hidden } \dots (0)\ 68] \\ \blacksquare (v - 1) \times 1000 &= 0.677\ 130\ 690 && [\text{Hidden } \dots (0)\ 69] \end{aligned}$$

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Numbers and Their Encodings

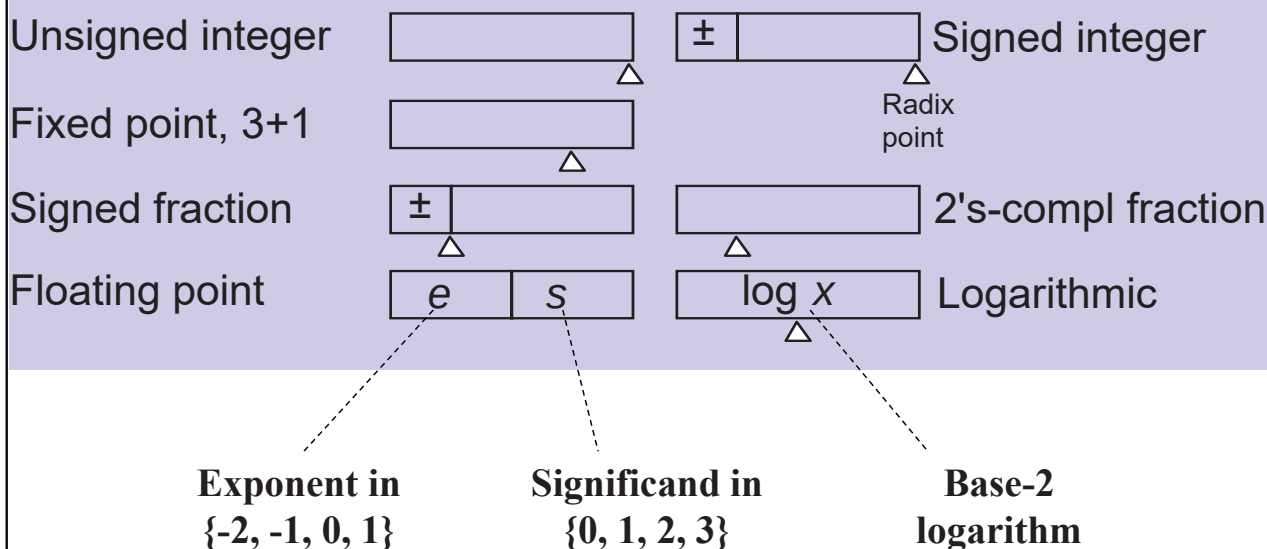


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Numbers and Their Encodings (cont'd)

- Formats of unsigned integer, fixed-point, signed fraction and floating-point



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Fixed-Radix Positional Number Systems

- A positive number with radix- r

$$(x_{K-1}x_{K-2} \dots x_1x_0 \cdot x_{-1}x_{-2} \dots x_{-L})_r = \sum_{i=-L}^{K-1} x_i r^i$$

- Fractional radix number system

- $r=10$ with digit set $[0, 9]$

$$\begin{aligned} (x_{K-1}x_{K-2} \dots x_1x_0 \cdot x_{-1}x_{-2} \dots x_{-L})_{\frac{1}{10}} &= \sum_i x_i 10^{-i} \\ &= (x_{-L} \dots x_{-2}x_{-1}x_0 \cdot x_1x_2 \dots x_{K-1})_{10} \end{aligned}$$

Signed-Magnitude Representation

- A 4-bit signed-magnitude number representation system for integers

- 1-bit for signed
 - 0: for positive
 - 1 for negative numbers

- 3-bit for magnitude

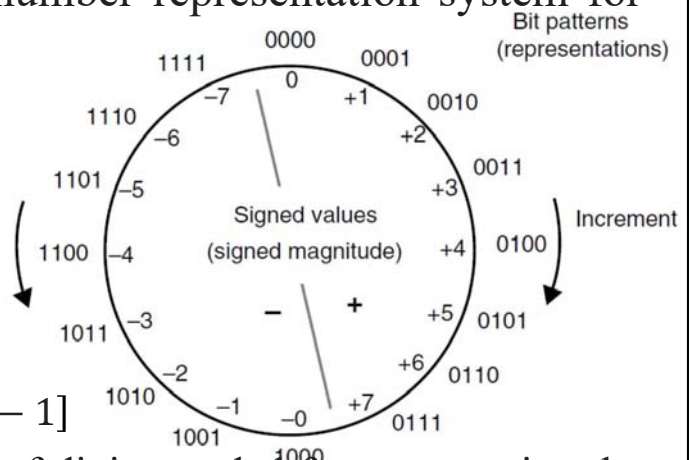
- Two zeros: +0 and -0

- Range: $[-(2^{K-1} - 1), 2^{K-1} - 1]$

- For radix- r , the number K of digits needed for representing the natural numbers up to max as

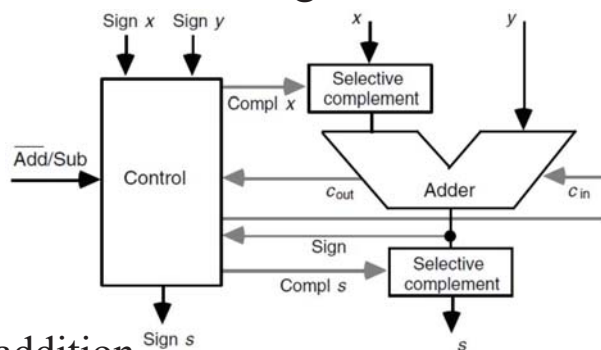
$$K = \lceil \log_r \max \rceil + 1 = \lceil \log_r (\max + 1) \rceil.$$

- The interval $[0, r^K - 1] = [0, r^K)$ of natural numbers



Disadvantage of the Signed-Magnitude Representation

- ❑ Operation may depend on the signs of the operands
- ❑ Example: adding a positive number X and a negative number $-Y$:
 - $X+(-Y)$
 - If $Y>X$, final result is $-(Y-X)$
- ❑ Calculation
 - Switch order of operands
 - Perform subtraction rather than addition
 - Attach the minus sign
- ❑ A sequence of decisions must be made, costing excess control logic and execution time.
- ❑ Avoid in the complement representation methods

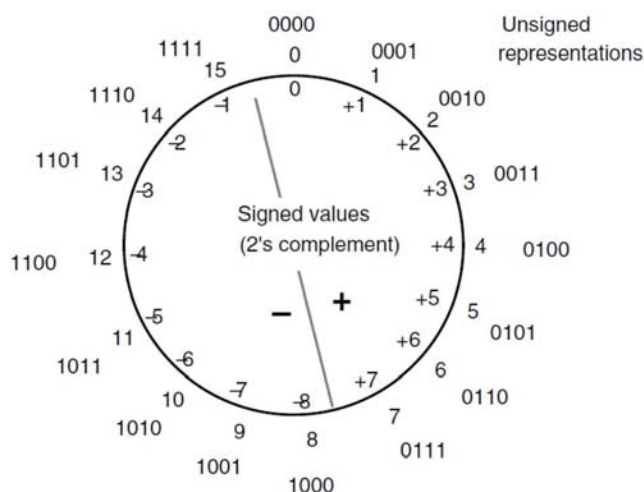


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2's- and 1's-Complement Numbers

- ❑ A 4-bit 2's-complement number representation system for integers in $[-N, +P]$ ($M = N + P + 1$)
 - 4-bit 2's-complement integer
 - $K = 4; M = 2^K = 16$
 - Signed in MSB
 - Range $[-2^{K-1}, 2^{K-1} - 1]$



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2's- and 1's-Complement Numbers (cont'd)

□ Radix-complement representation

■ $M - X = r^K - X = \bar{X} + ulp$, where $ulp = 1$.

□ $X - Y = X + (2's \text{ Complement of } Y)$

$$= X + (M - Y) = X + (r^K - Y) = X + (\bar{Y} + 1)$$



Example - Addition in 2's Complement

□ Calculating $X + (-Y)$ with $Y > X \rightarrow 3 + (-5)$

$$\begin{array}{r} 0011 \quad 3 \\ + 1011 \quad -5 \\ \hline 1110 \quad -2 \end{array}$$

- Correct result represented in the 2's complement method - no need for preliminary decisions or post corrections

□ Calculating $X + (-Y)$ with $X > Y \rightarrow 5 + (-3)$

$$\begin{array}{r} 0101 \quad 5 \\ + 1101 \quad -3 \\ \hline 1 \ 0010 \quad 2 \end{array}$$

- Only the last four least significant digits are retained, yielding 0010

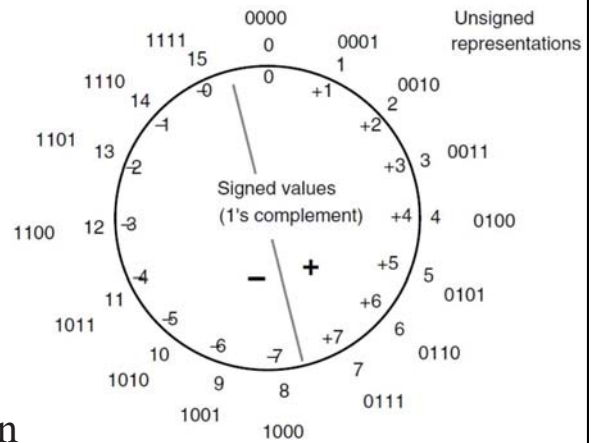
- Discard MSB

2's- and 1's-Complement Numbers (cont'd)

□ A 4-bit 1's-complement number representation system for integers in $[-N, +P]$ ($N = P$)

■ 4-bit 2's-complement integer

- $K = 4; M = 2^K - 1 = 15$
- Signed in MSB
- Range $[-(2^{K-1} - 1), 2^{K-1} - 1]$

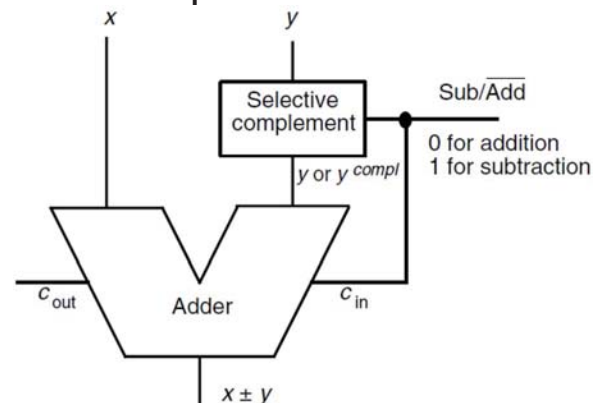


■ Radix-complement representation

- $M - X = (r^K - ulp) - X = \bar{X}$, where $ulp = 1$

2's- and 1's-Complement Numbers (cont'd)

□ Adder/subtractor architecture for 2's complement numbers





Addition/Subtraction in Complement Methods

- Example - (2's complement)

$$\begin{array}{r} 01001 \ 9 \\ + 11001 \ -7 \\ \hline 1 \ 00010 \ 2 \end{array}$$

- Carry-out **discarded** - does not indicate **overflow**

- In general, if X and Y have **opposite signs** - no overflow can occur regardless of whether there is a carry-out or not.

- Examples - (2's complement)

$$\begin{array}{r} 0 \ 0 \ 1 \ 0 \ 1 \ 5 \\ + 1 \ 0 \ 1 \ 1 \ 0 \ -10 \\ \hline 1 \ 1 \ 0 \ 1 \ 1 \ -5 \end{array} \quad \text{No carry-out}$$

$$\begin{array}{r} 0 \ 1 \ 0 \ 1 \ 0 \ 10 \\ + 1 \ 1 \ 0 \ 1 \ 1 \ -5 \\ \hline 1 \ 0 \ 0 \ 1 \ 0 \ 1 \ 5 \end{array} \quad \text{Carry-out}$$

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Addition/Subtraction - Complement (cont'd)

- If X and Y have the **same sign** and result has different sign.

■ Overflow

- Examples - (2's complement)

$$\begin{array}{r} 10111 \ -9 \\ + 10111 \ -9 \\ \hline 1 \ 01110 \ 14 = -18 \text{ mod } 32 \end{array}$$

- Carry-out and overflow

$$\begin{array}{r} 01001 \ 9 \\ + 00111 \ 7 \\ \hline 0 \ 10000 \ -16 = 16 \text{ mod } 32 \end{array}$$

- No carry-out but overflow

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Shift Operations for SM, 2's- and 1's Complement

- ❑ Signed-magnitude numbers
 - ... 0 0 0 { $x_{K-2} \dots x_1 x_0 x_{-1} x_{-2} \dots x_{-L}$ } 0 0 0 ...
 - $x_{K-2} \dots x_{-L}$: the magnitude part
- ❑ $(K + L)$ bits 2's and 1's complement numbers
 - K : Integer bit numbers
 - L : Fractional bit numbers

Range/precision extension for 2's-complement numbers

$\dots x_{K-1} \quad x_{K-1} \quad x_{K-1} \quad | \quad x_{K-1} \quad x_{K-2} \dots x_1 \quad x_0 \quad x_{-1} \quad x_{-2} \dots x_{-L} \quad | \quad 0 \quad 0 \quad 0 \dots$

← Sign extension → Signed-bit LSB ← Extension →

MSB

Range/precision extension for 1's-complement numbers

$\dots x_{K-1} \quad x_{K-1} \quad x_{K-1} \quad x_{K-1} \quad x_{K-2} \quad \dots \quad x_1 \quad x_0 \cdot x_{-1} \quad x_{-2} \quad \dots \quad x_{-L}$

← Sign extension → Signed-bit LSB ← Extension →

MSB

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Shift Operations for 2's- and 1's Complement

- Example
 - 2's complement
$$\text{SH.L}\{00110=6\}=0110\textcolor{red}{0}=12$$
$$\text{SH.R}\{00110=6\}=\textcolor{red}{0}0011=3$$
$$\text{SH.L}\{11010=-6\}=1010\textcolor{red}{0}=-12$$
$$\text{SH.R}\{11010=-6\}=\textcolor{red}{1}1101=-3$$
 - 1's complement
$$\text{SH.L}\{11001=-6\}=1001\textcolor{red}{1}=-12$$
$$\text{SH.R}\{11001=-6\}=\textcolor{red}{1}1100=-3$$

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SM, 2's- and 1's-Complement Representation

Sequence	Two's complement	One's complement	Signed-magnitude
0111	7	7	7
0110	6	6	6
0101	5	5	5
0100	4	4	4
0011	3	3	3
0010	2	2	2
0001	1	1	1
0000	0	0	0
1111	-1	-0	-7
1110	-2	-1	-6
1101	-3	-2	-5
1100	-4	-3	-4
1011	-5	-4	-3
1010	-6	-5	-2
1001	-7	-6	-1
1000	-8	-7	-0

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Floating-Point Numbers

❑ Various systems can be used for a subset of real numbers.

- Fixed-point $\pm w.f$ (Low precision and/or range)
- Rational $\pm p/q$ (Difficult arithmetic)
- Floating-point $\pm s \times b^e$ (Most common scheme)
- Logarithmic $\pm \log_b x$ (Limiting case of floating-point)

❑ Fixed-point numbers

- $x = (0000\ 0000 . 0000\ 1001)_2$ Small number
- $y = (1001\ 0000 . 0000\ 0000)_2$ Large number

❑ Floating-point numbers

- $x = \pm s \times b^e$ or $\pm \text{significand} \times \text{base}^{\text{exponent}}$

$$x = 1.001 \times 2^{-5}$$

$$y = 1.001 \times 2^{+7}$$

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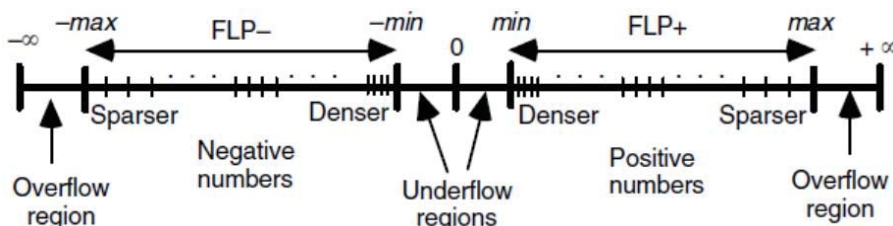
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Floating-Point Number Format and Distribution

Typical floating-point (FLP) number format

$\pm$	e	s
Sign	Exponent Signed integer, 0 : + often represented 1 : - as unsigned value by adding a bias.	Significand Represented as a fixed-point number Usually normalized by shifting, so that the MSB becomes nonzero. In radix 2, the fixed leading 1 can be removed to save 1 bit; this bit is known as "hidden 1."
	Range with h bits: $[-bias, 2^h - 1 - bias]$.	

Subranges and special values in floating-point (FLP) number representations



- $|max| = \text{Largest significand} \times 2^{\text{Largest Exponent}}$
- $|min| = \text{Smallest significand} \times 2^{\text{Smallest Exponent}}$
- Three special values: $-\infty, 0, \infty$

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IEEE Floating-Point Standard

IEEE 754-2008

Sign Biased exponent Significand $s = 1.f$ (the 1 is hidden)

$\pm$	$e + \text{bias}$	f
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16-bit:	5 bits, bias = 15	10 + 1 bits, half precision format
32-bit:	8 bits, bias = 127	23 + 1 bits, single-precision or short format
64-bit:	11 bits, bias = 1023	52 + 1 bits, double-precision or long format
128-bit:	15 bits, bias = 16383	112 + 1 bits, quadruple-precision format

Single precision format

1-bit 8-bit 23-bit

Sign	Exponent	Fraction (mantissa)
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$b_{31}, b_{30}, \dots, b_{23}, b_{22}, \dots, b_0$

The real value can be expressed as

$$(-1)^{\text{Sign}} \times 1.\text{Fraction} \times 2^{\text{Exponent}-127}$$

$$= (-1^{b_{31}}) \times (1.0 + b_{22} \times 2^{-1} + b_{21} \times 2^{-2} + \dots + b_0 \times 2^{-23}) \times 2^{\text{Exponent}-127}$$

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Some Features of the IEEE 754-2008 Short and Long FLP

Feature	Single/Short	Double/Long
Word width, bits	32	64
Significand bits	23 + 1 hidden	52 + 1 hidden
Significand range	$[1, 2 - 2^{-23}]$	$[1, 2 - 2^{-52}]$
Exponent bits	8	11
Exponent bias	127	1023
Zero (± 0)	$e + bias = 0, f = 0$	$e + bias = 0, f = 0$
Subnormal	$e + bias = 0, f \neq 0$ represents $\pm 0.f \times 2^{-126}$	$e + bias = 0, f \neq 0$ represents $\pm 0.f \times 2^{-1022}$
Infinity ($\pm \infty$)	$e + bias = 255, f = 0$	$e + bias = 2047, f = 0$
Not-a-number (NaN)	$e + bias = 255, f \neq 0$	$e + bias = 2047, f \neq 0$
Ordinary number	$e + bias \in [1, 254]$ $e \in [-126, 127]$ represents $1.f \times 2^e$	$e + bias \in [1, 2046]$ $e \in [-1022, 1023]$ represents $1.f \times 2^e$
<i>min</i>	$2^{-126} \approx 1.2 \times 10^{-38}$	$2^{-1022} \approx 2.2 \times 10^{-308}$
<i>max</i>	$\approx 2^{128} \approx 3.4 \times 10^{38}$	$\approx 2^{1024} \approx 1.8 \times 10^{308}$

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Signed Digit (SD) Representation

- A large number of consecutive 1s can be simplified based on the following fact.

- $2^j + 2^{j-1} + \dots + 2^{i+1} + 2^i = 2^{j+1} - 2^i$

- For example, $1111_2 = 1000\bar{1}_2 (= 2^4 - 2^0)$, where $\bar{1} = -1$.
 $15_d = 16_d - 1_d$

- Radix-2 Booth's recoding

- x_i : Encoding bit

- x_{i-1} : Reference bit

- y_i : Coded bit

x_i	x_{i-1}	y_i	Explanation
0	0	0	No string of 1s in sight
0	1	1	End of string of 1s in x
1	0	$\bar{1}$	Beginning of string of 1s in x
1	1	0	Continuation of string of 1s in x

- Example:

$$(1) \begin{array}{cccc} 1001 & 1101 & 1010 & 1110 \\ \bar{1}010 & 0\bar{1}10 & \bar{1}1\bar{1}1 & 00\bar{1}0 \end{array}$$

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SD Representation of Twiddle Factors

□ Twiddle factor

$$W_N^{nk} = e^{-j2\pi nk/N}$$

TWF		SD Expression			Floating-Point Value	SQNR (dB)
		SD Code	Shift-and-Add Operation	Fixed-Point Value		
W_{64}^0	Re	1000000000	2^0	1.0	1.0000000000000000	--
	Im	0000000000	--	0.0	0.0000000000000000	--
W_{64}^1	Re	1000000110	$2^0 - 2^{-7} + 2^{-8}$	0.996093750	0.995184726672197	60.79
	Im	0001010010	$2^{-3} - 2^{-5} + 2^{-8}$	0.097656250	0.098017140329561	48.68
W_{64}^2	Re	1000010110	$2^0 - 2^{-5} + 2^{-7} + 2^{-8}$	0.980468750	0.980785280403230	69.82
	Im	0001100100	$2^{-3} + 2^{-4} + 2^{-7}$	0.195312500	0.195090322016128	58.87
W_{64}^3	Re	1000101010	$2^0 - 2^{-4} + 2^{-6} + 2^{-8}$	0.957031250	0.956940335732209	80.45
	Im	0010010100	$2^{-2} + 2^{-5} + 2^{-7}$	0.289062500	0.290284677254462	47.51
W_{64}^4	Re	1001011001	$2^0 - 2^{-3} + 2^{-5} + 2^{-6} + 2^{-9}$	0.923828125	0.923879532511287	85.09
	Im	0101000100	$2^{-1} - 2^{-3} + 2^{-7}$	0.382812500	0.382683432365090	69.44
W_{64}^5	Re	1001000100	$2^0 - 2^{-3} + 2^{-7}$	0.882812500	0.881921264348355	59.91
	Im	0100010010	$2^{-1} - 2^{-5} + 2^{-8}$	0.472656250	0.471396736825998	51.46
W_{64}^6	Re	0110101010	$2^{-1} + 2^{-2} + 2^{-4} + 2^{-6} + 2^{-8}$	0.832031250	0.831469612302545	63.41
	Im	0100100100	$2^{-1} + 2^{-4} - 2^{-7}$	0.554687500	0.555570233019602	55.98
W_{64}^7	Re	0110010100	$2^{-1} + 2^{-2} + 2^{-5} - 2^{-7}$	0.773437500	0.773010453362737	65.15
	Im	0101000101	$2^{-1} + 2^{-3} + 2^{-7} + 2^{-9}$	0.634765625	0.634393284163645	64.63
W_{64}^8	Re	0101101010	$2^{-1} + 2^{-3} + 2^{-4} + 2^{-6} + 2^{-8}$	0.707031250	0.707106781186548	79.43
	Im	0101101010	$2^{-1} + 2^{-3} + 2^{-4} + 2^{-6} + 2^{-8}$	0.707031250	0.707106781186548	79.43
Average SQNR						65.00

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